FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry | Unit 10 — Practical Geometry

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 1**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question (shortened)	Answer	Explanation
1	Which number is irrational?	A	**√7** is irrational (7 is not a perfect square). √16 = 4, 22/7, and $0.\overline{7}$ = 7/9 are all rational (they can be written as a ratio of integers).
2	Simplest form of √50	B	**5√2**, because √50 = √(25 × 2) = √25 × √2 = 5√2.
3	$\log_{10} 1000$	C	**3**, because $10^3 = 1000$, so $\log_{10} 1000 = 3$.
4	log(mn) = ?	B	**log m + log n** (product law of logarithms). The log of a product equals the sum of the logs.
5	Factors of $x^2 - 7x + 12$	A	**(x − 3)(x − 4)**, since (−3) + (−4) = −7 and (−3)(−4) = +12.
6	Factorization of $a^3 + 8$	D	**$(a + 2)(a^2 - 2a + 4)$**. Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with b = 2.
7	Solve 2x + 3 = 11	C	**x = 4**, since 2x = 11 − 3 = 8, so x = 8/2 = 4.
8	Solution set of |x| = 5	D	**{−5, 5}**. The absolute value equation |x| = 5 has two solutions: x = 5 and x = −5.
9	Distance (0,0) to (3,4)	B	**5**, because $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
10	Midpoint of (2,4) & (6,10)	D	**(4, 7)**, using ((2+6)/2, (4+10)/2) = (8/2, 14/2) = (4, 7).
11	Centre to a point on circle	A	**Radius.** A chord joins two points on the circle; a diameter is a chord through the centre; a tangent touches the circle at exactly one point.
12	Measurements for a triangle	C	**3.** A unique triangle is fixed by three independent elements (e.g., SSS, SAS, ASA).

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) Simplify √72; rationalize 3/√2 **3 marks**

(a) √72 = √(36 × 2) = 6√2. **1.5 marks**

(b) 3/√2 = (3 × √2) / (√2 × √2) = 3√2 / 2. **1.5 marks**

OR

Rational or irrational? **3 marks**

- (a) √36 = 6 → **rational** (a perfect square). **1 mark**
- (b) √10 → **irrational** (10 is not a perfect square; the decimal is non-terminating, non-recurring). **1 mark**
- (c) $0.\overline{7}$ = 7/9 → **rational** (a recurring decimal can be written as a fraction). **1 mark**

2(ii) Laws of exponents **3 marks**

(a) $(3^2 \times 3^4) \div 3^3 = 3^{(2+4-3)} = 3^3 = \mathbf{27}$. **1.5 marks**

(b) $(a^{-2})^{-3} = a^{(-2)(-3)} = a^6$. **1.5 marks**

OR

Scientific notation **3 marks**

- (a) $0.00046 = 4.6 \times 10^{-4}$. **1.5 marks**
- (b) $92{,}000{,}000 = 9.2 \times 10^7$. **1.5 marks**

2(iii) Evaluate logarithms 3 marks

(a) $\log_2 32 = \log_2 2^5 = \mathbf{5}$. 1.5 marks

(b) $\log_{10} 1000 - \log_{10} 10 = 3 - 1 = \mathbf{2}$. 1.5 marks

OR

Expand log(a²b / c) 3 marks

$\log(a^2b / c) = \log(a^2b) - \log c = \log a^2 + \log b - \log c = \mathbf{2 \log a + \log b - \log c}$.
1 mark quotient law, 1 mark product law, 1 mark power law.

2(iv) Using log 2 = 0.3010, log 3 = 0.4771 3 marks

(a) $\log 6 = \log(2 \times 3) = \log 2 + \log 3 = 0.3010 + 0.4771 = \mathbf{0.7781}$. 1.5 marks

(b) $\log 12 = \log(2^2 \times 3) = 2 \log 2 + \log 3 = 0.6020 + 0.4771 = \mathbf{1.0791}$. 1.5 marks

OR

Find x 3 marks

- (a) $\log_3 x = 4 \to x = 3^4 = \mathbf{81}$. 1.5 marks
- (b) $\log_x 81 = 4 \to x^4 = 81 \to x = \mathbf{3}$. 1.5 marks

2(v) Factorize 3 marks

(a) $x^2 - 9x + 20 = \mathbf{(x - 4)(x - 5)}$ $[-4 - 5 = -9; (-4)(-5) = 20]$. 1.5 marks

(b) $2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = \mathbf{(2x + 1)(x + 3)}$. 1.5 marks

OR

Factorize (sum/difference of cubes) 3 marks

(a) $8a^3 + 27 = (2a)^3 + 3^3 = \mathbf{(2a + 3)(4a^2 - 6a + 9)}$. 1.5 marks

(b) $64x^3 - 1 = (4x)^3 - 1^3 = \mathbf{(4x - 1)(16x^2 + 4x + 1)}$. 1.5 marks

2(vi) HCF of $12x^2y^3$ and $18x^3y^2$ 3 marks

$12x^2y^3 = 2^2 \cdot 3 \cdot x^2 \cdot y^3$; $18x^3y^2 = 2 \cdot 3^2 \cdot x^3 \cdot y^2$. 1.5 marks

HCF = (common factors with lowest powers) = $2 \cdot 3 \cdot x^2 \cdot y^2 = \mathbf{6x^2y^2}$. 1.5 marks

OR

LCM of $x^2 - 1$ and $x^2 + 2x + 1$ 3 marks

$x^2 - 1 = (x - 1)(x + 1)$; $x^2 + 2x + 1 = (x + 1)^2$. 1.5 marks

LCM = (highest powers of all factors) = $\mathbf{(x - 1)(x + 1)^2}$. 1.5 marks

2(vii) Solve (2x)/3 + 1 = x/2 + 2 3 marks

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Multiply both sides by 6 (LCD):
6·(2x/3) + 6·1 = 6·(x/2) + 6·2
4x + 6 = 3x + 12
4x − 3x = 12 − 6 → x = 6
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1 mark clearing fractions, 1 mark simplifying, 1 mark answer.

OR

Solve |2x + 5| = 11 3 marks

$2x + 5 = 11 \to 2x = 6 \to x = 3$. OR $2x + 5 = -11 \to 2x = -16 \to x = -8$.
Solution set = $\mathbf{\{3, -8\}}$. 1.5 marks each case.

2(viii) Solve 8x + 9 < 6x − 7 3 marks

$8x - 6x < -7 - 9 \to 2x < -16 \to x < -8$. 2 marks
Solution set = $\mathbf{\{ x \mid x < -8, x \in \mathbb{R} \}}$. 1 mark

OR

Solve 2(x − 3) ≥ 4x + 2 3 marks

$2x - 6 \geq 4x + 2 \to -6 - 2 \geq 4x - 2x \to -8 \geq 2x \to x \leq -4$. 2 marks
Solution set = $\mathbf{\{ x \mid x \leq -4, x \in \mathbb{R} \}}$. 1 mark

2(ix) Distance A(−2, 1), B(1, 5) 3 marks

$d = \sqrt{[(1 - (-2))^2 + (5 - 1)^2]} = \sqrt{[(3)^2 + (4)^2]} = \sqrt{(9 + 16)} = \sqrt{25} =$ **5 units**.

1 mark formula, 1 mark substitution, 1 mark answer.

OR

Distance P(3, −2), Q(−1, 1) 3 marks

$d = \sqrt{[(-1 - 3)^2 + (1 - (-2))^2]} = \sqrt{[(-4)^2 + (3)^2]} = \sqrt{(16 + 9)} = \sqrt{25} =$ **5 units**. 3 marks

2(x) Midpoint of (2, 5) & (6, 9) 3 marks

Midpoint = $((2 + 6)/2, (5 + 9)/2) = (8/2, 14/2) =$ **(4, 7)**. 2 marks

Both coordinates are positive, so the midpoint lies in **Quadrant I**. 1 mark

OR

Find a and b from midpoint M(1, 4) 3 marks

$(6 + a)/2 = 1 \rightarrow 6 + a = 2 \rightarrow a =$ **−4**. 1.5 marks

$(b + 2)/2 = 4 \rightarrow b + 2 = 8 \rightarrow b =$ **6**. 1.5 marks

2(xi) Practical geometry terms 3 marks

- **Chord:** a line segment whose two endpoints both lie on the circle. 1 mark
- **Tangent:** a straight line that touches the circle at exactly one point (the point of contact). 1 mark
- **Arc:** a continuous part of the circumference (curved boundary) of the circle. 1 mark

OR

Steps to bisect a line segment AB 3 marks

- With centre A and a radius more than half of AB, draw arcs above and below AB. 1 mark
- With the same radius and centre B, draw two more arcs to cut the first arcs at points P and Q. 1 mark
- Draw the straight line PQ; it cuts AB at its midpoint M and is the perpendicular bisector of AB. 1 mark

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

3(i) Factorize product − 24; square root by factorization 5 marks

(a) $(x + 1)(x + 2)(x + 3)(x + 4) - 24$: 3 marks

Pair the factors: $[(x+1)(x+4)]\ [(x+2)(x+3)] - 24$

$= (x^2 + 5x + 4)(x^2 + 5x + 6) - 24$

Let $y = x^2 + 5x$: $(y + 4)(y + 6) - 24$

$= y^2 + 10y + 24 - 24 = y^2 + 10y = y(y + 10)$

$= \mathbf{x(x + 5)(x^2 + 5x + 10)}$

(b) Square root of $4x^2 + 12xy + 9y^2$: 2 marks

$4x^2 + 12xy + 9y^2 = (2x + 3y)^2$, so the square root is **$\pm(2x + 3y)$**.

OR

Square root by division; HCF by factorization 5 marks

(a) $\sqrt{(x^4 + 8x^3 + 20x^2 + 16x + 4)}$: 3 marks

By the division method the square root is **$x^2 + 4x + 2$**.

Check: $(x^2 + 4x + 2)^2$

$= x^4 + 8x^3 + (16 + 4)x^2 + 16x + 4$

$= x^4 + 8x^3 + 20x^2 + 16x + 4$ ✓

(b) HCF of $x^2 - 5x + 6$ and $x^2 - 9$: 2 marks

$x^2 - 5x + 6 = (x - 2)(x - 3)$; $x^2 - 9 = (x - 3)(x + 3)$. HCF = **$(x - 3)$**.

3(ii) Laws of logarithms; evaluation [5 marks]

(a) Any three laws of logarithms: [2 marks]

- Product law: $\log(mn) = \log m + \log n$.
- Quotient law: $\log(m/n) = \log m - \log n$.
- Power law: $\log(m^n) = n \log m$. (Also valid: $\log_a a = 1$, $\log_a 1 = 0$.)

(b) Using log 2 = 0.3010, log 3 = 0.4771, log 7 = 0.8451: [3 marks]

- (i) $\log 42 = \log(2 \times 3 \times 7) = 0.3010 + 0.4771 + 0.8451 = \mathbf{1.6232}$. [1.5 marks]
- (ii) $\log(9/2) = \log 9 - \log 2 = 2 \log 3 - \log 2 = 0.9542 - 0.3010 = \mathbf{0.6532}$. [1.5 marks]

OR

Definitions; find x [5 marks]

(a) Definitions: [2 marks]

- **Common logarithm:** a logarithm to base 10, written log x.
- **Natural logarithm:** a logarithm to base e ($e \approx 2.718$), written ln x.
- **Characteristic:** the integral (whole-number) part of a logarithm; **Mantissa:** the positive decimal (fractional) part.

(b) Find x: [3 marks (1 each)]

(i) $\log_{10} x = 2 \rightarrow x = 10^2 = \mathbf{100}$

(ii) $\log_2(x - 1) = 3 \rightarrow x - 1 = 2^3 = 8 \rightarrow x = \mathbf{9}$

(iii) $\log_5 125 = x \rightarrow 5^x = 125 = 5^3 \rightarrow x = \mathbf{3}$

3(iii) Linear equation and inequalities [5 marks]

(a) Solve (x − 2)/3 + (x + 1)/2 = 4: [3 marks]

Multiply by 6 (LCD): $2(x - 2) + 3(x + 1) = 24$

$2x - 4 + 3x + 3 = 24 \rightarrow 5x - 1 = 24$

$5x = 25 \rightarrow x = \mathbf{5}$

Verify: $(5-2)/3 + (5+1)/2 = 1 + 3 = 4$ ✓

(b) Solve 3x − 5 ≤ 7: [2 marks]

$3x \le 12 \rightarrow x \le 4$. Solution set = $\{ x \mid x \le 4, x \in \mathbb{R} \}$. On the number line, a filled (closed) circle at 4 with shading to the left.

OR

Absolute value equation; compound inequality [5 marks]

(a) Solve |3x − 2| = |x + 4|: [3 marks]

Case 1: $3x - 2 = x + 4 \rightarrow 2x = 6 \rightarrow x = 3$

Case 2: $3x - 2 = -(x + 4) \rightarrow 4x = -2 \rightarrow x = -1/2$

Solution set = $\{ 3, -1/2 \}$

(b) Solve −3 < 2x + 1 ≤ 7: [2 marks]

Subtract 1: $-4 < 2x \le 6$. Divide by 2: $-2 < x \le 3$. Solution set = $\{ x \mid -2 < x \le 3, x \in \mathbb{R} \}$.

3(iv) Vertices of a square; distance formula 5 marks

(a) Show A(−4, 2), B(1, 4), C(3, −1), D(−2, −3) form a square: 4 marks

$$AB = \sqrt{(1+4)^2 + (4-2)^2} = \sqrt{25+4} = \sqrt{29}$$
$$BC = \sqrt{(3-1)^2 + (-1-4)^2} = \sqrt{4+25} = \sqrt{29}$$
$$CD = \sqrt{(-2-3)^2 + (-3+1)^2} = \sqrt{25+4} = \sqrt{29}$$
$$DA = \sqrt{(-4+2)^2 + (2+3)^2} = \sqrt{4+25} = \sqrt{29}$$

All four sides equal ($\sqrt{29}$). The diagonals: $AC = \sqrt{(3+4)^2 + (-1-2)^2} = \sqrt{58}$ and $BD = \sqrt{(-2-1)^2 + (-3-4)^2} = \sqrt{58}$. Since all sides are equal **and** the diagonals are equal, ABCD is a **square**.

(b) Distance formula: 1 mark

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

OR

Constructions 5 marks

(a) Triangle ABC, AB = 5 cm, BC = 6 cm, ∠B = 60°: 3 marks

- Draw base BC = 6 cm with a ruler. 1 mark
- At B, construct an angle of 60° using a compass (draw an arc, then the 60° arc) and draw ray BX. 1 mark
- From B, cut BA = 5 cm along BX; join A to C. Triangle ABC is the required triangle. 1 mark

(b) Perpendicular bisector of a 6 cm segment: 2 marks

- Draw segment PQ = 6 cm. With centres P and Q and radius more than 3 cm, draw arcs on both sides that intersect at two points. 1 mark
- Join the two intersection points; this line is perpendicular to PQ and passes through its midpoint (3 cm from each end). 1 mark

— End of Answer Key —